Yingjing **Huang**

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Research Interest

GIScience, GeoAI, Deep Learning, Computer Vision, Urban Visual Intelligence, Social Sensing

Experience

University of Vienna — *Vienna, Austria*
Postdoc — *Mar. 2026 - current*
- Advisor: Krzysztof Janowicz

Peking University — *Beijing, China*
Research Associate — *Jul. 2025 - Feb. 2026*
- Advisor: Fan Zhang

MIT Senseable City Lab — *Cambridge, US*
Visiting Ph.D. Student — *Jan. 2023 - Jan. 2024*
- Advisors: Carlo Ratti, Fábio Duarte, Simone Mora

Education

Peking University — *Beijing, China*
Ph.D. in Geographic Information Science — *Sep. 2021 - Jun. 2025*
- Advisors: Yu Liu, Fan Zhang

Wuhan University — *Wuhan, China*
MS. in Geographic Information Science — *Sep. 2018 - Jun. 2021*
- Advisor: Teng Fei

Wuhan University — *Wuhan, China*
BSc. in Geographic Science — *Sep. 2013 - Jun. 2017*
- Advisors: Yiyun Chen, Teng Fei

Publications

Publications reported in reverse chronological order
(# co-first author, Google citations = 262)

Ongoing and Submitted Manuscripts

[1] **Yingjing Huang**#, Yong Li#, Fan Zhang, & Yu Liu. Measuring sustainable development with urban imagery. (In Preparing)

[2] **Yingjing Huang**, Yuqing Wang, Ce Hou, Yanhua Yao, Fabio Duarte, & Fan Zhang. 2025. Revealing urban inequality from a dual perspective of indoor and outdoor spaces. *Annals of the American Association of Geographers.* (Under Review)

[3] Jiapeng Li, **Yingjing Huang**, Fan Zhang, Yu Liu. 2026. StreetTree: A Large-Scale Global Benchmark for Fine-Grained Tree Species Classification. *Arxiv:2602.19123.* (Under Review)

Peer-reviewed Journal Articles

[1] Yong Li#, **Yingjing Huang**#, Fan Zhang. 2026. Learning street view representations based on a spatiotemporal contrastive learning framework. *Computers, Environment and Urban Systems.* (In Printing, IF = 8.3, Rank in Urban Studies = 3/288, Q1)

[2] Jing Xiao, Teng Fei, Bo Yu, **Yingjing Huang**, & Yunyan Du. 2025. Street view images help to reveal the impact of noisy environments on the survival duration of stroke patients. International Journal of

Health Geographics, 24(1), 27. (IF = 3.2, Rank in Public Health, Environmental and Occupational Health = 106/687, Q1)

[3] Yu Liu, Xuechen Wang, Yidan Wang, Fei Huang, **Yingjing Huang**, Yong Li, Weiyu Zhang, Shuhui Gong, Gengchen Mai, Yao Yao, Yang Yue, Haifeng Li, & Fan Zhang. 2025. Representation learning for geospatial data. *Annals of GIS*, 1–27. (IF = 3.3, Rank in Earth and Planetary Sciences = 25/198, Q1)

[4] Jin Zhu, **Yingjing Huang**, Ziyue Cao, Yue Zhang, Yuan Ding, & Jinglong Du. 2025. Evaluating urban greenery through the front-facing street view imagery: Insights from a Nanjing Case Study. *ISPRS International Journal of Geo-Information*, 14(8), 287. (IF = 2.8, Rank in Geography, Planning and Development = 95/841, Q2)

[5] **Yingjing Huang**, Fan Zhang, Lun Wu, and Yu Liu. 2025. Measuring urban physical environments using image deep features. *Cities*, 166, 106196. (IF = 6.6, Rank in Urban Studies = 7/288, Q1)

[6] **Yingjing Huang**, Fan Zhang, Yong Li, Lun Wu, & Yu Liu. 2025. Intelligent computational representation of urban imagery. *Geomatics and Information Science of Wuhan University.* (In Chinese, a leading journal in Geomatics in China, indexed by EI Compendex)

[7] **Yingjing Huang**, Rohit Sanatani, Chang Liu, Yuhao Kang, Fan Zhang, Yu Liu, Fabio Duarte, and Carlo Ratti. (2025). No "true" greenery: Deciphering the bias of satellite and street view imagery in urban greenery measurement. *Building and Environment*, 269, 112395. (IF = 7.6, Rank in Geography, Planning and Development = 18/841, Q1)

[8] **Yingjing Huang**, Fan Zhang, Yong Gao, Wei Tu, Fabio Duarte, Carlo Ratti, Diansheng Guo, and Yu Liu. 2023. Comprehensive urban space representation with varying numbers of street-level images. *Computers, Environment and Urban Systems*, 106, 102043. (IF = 8.3, Rank in Urban Studies = 3/288, Q1)

[9] Meng Bian, Shuyi Guo, Wei Wang, Yuhui Ouyang, **Yingjing Huang**, and Teng Fei. 2021. Nextday forecast of Beijing pollen concentration fused with vegetation remote sensing data——Implementing a nonlinear autoregressive neural network model with external input. *Journal of Geo-Information Science*, 2021,23(09):1705-1713. (in Chinese, a key academic journal in GIS, indexed in the CSCD Core database)

[10] **Yingjing Huang**, Teng Fei, Mei-Po Kwan, Yuhao Kang, Jun Li, Yizhuo Li, Xiang Li, and Meng Bian. 2020. GIS-based emotional computing: A review of quantitative approaches to measure the emotion layer of human–environment relationships. *ISPRS International Journal of Geo Information*, 9, 551. (IF = 2.8, Rank in Geography, Planning and Development = 95/841, Q2)

[11] **Yingjing Huang**, Jun Li, Guofeng Wu and Teng Fei. 2020. Quantifying the bias in place emotion extracted from photos on social networking sites: A case study on a university campus. *Cities*, 102, 102719. (IF = 6.6, Rank in Urban Studies = 7/288, Q1)

[12] Yizhuo Li, Teng Fei, **Yingjing Huang**, Jun Li, Xiang Li, Fan Zhang, Yuhao Kang and Guofeng Wu. 2020. Emotional habitat: Mapping the global geographic distribution of human emotion with physical environmental factors using a species distribution model. *International Journal of Geographical Information Science*, 1-23. (IF = 5.1, Rank in Geography, Planning and Development = 31/841, Q1)

[13] Shuangyin Zhang, Jun Li, Siying Wang, **Yingjing Huang**, Yizhuo Li, Yiyun Chen and Teng Fei. 2020. Repaid identification and prediction of cadmium–lead cross-stress of different stress levels in rice canopy based on visible and near-infrared spectroscopy. *Remote Sensing*, 12, 469. (IF = 4.1, Rank in General Earth and Planetary Sciences = 13/198, Q1)

[14] Yiyun Chen, Tianci Qi, **Yingjing Huang**, Yuan Wan, Ruiying Zhao, Lin Qi, Chao Zhang, and Teng Fei. 2017. Optimization method of calibration dataset for VIS-NIR spectral inversion model of soil organic matter content. *Transactions of the Chinese Society of Agricultural Engineering*, 06, 107-114. (in Chinese, a top-tier journal in Agricultural Engineering in China, indexed by EI Compendex)

Patents

[1] Fan Zhang, **Yingjing Huang**, Yu Liu. Method, Device, and Storage Medium for Comprehensive Landscape Representation Based on Street Imagery. China Patent ZL202510142438.1, Granted December

16, 2025.

[2] Fan Zhang, **Yingjing Huang**, Diansheng Guo. Image Processing Method, Device, Product, Equipment, and Medium. China Patent Application CN116977690A, Published October 31, 2023, Patent Pending.

[3] Fan Zhang, **Yingjing Huang**, Diansheng Guo. Method and Related Device for Identifying Regions in Space. China Patent Application CN116958803A, Published October 27, 2023, Patent Pending.

Honors & Awards

2025	**Postdoc-NeT-AI Fellowship on Explainable AI** A highly competitive fellowship to conduct one week of visiting research in Germany, funded by German Academic Exchange Service (DAAD)	*Germany*
2022	**Visiting Doctoral Research Scholarship** A highly competitive fellowship to conduct one year of visiting research abroad, funded by China Scholarship Council (CSC)	*China*
2020	**"Excellent graduate student"** National award for the excellent graduate student in Wuhan University	*China*
2020	**Kwang-Hua Scholarship of Wuhan University** Scholarship to the excellent student	*China*
2020	**Second-Class Scholarship of Wuhan University** Awards to outstanding student in academic in Wuhan University	*China*
2015	**Third-Class Scholarship of Wuhan University** Awards to outstanding student in academic in Wuhan University	*China*

Talks

Invited talks

Analysis methods and applications of street view imagery *Online*
Nanjing Normal University, China *Aug. 2025*
[Link] (in Chinese)

Intelligent computational representation of urban imagery *Online*
GISChat (A non-profit, GIS knowledge-sharing platform organized by Chinese scholars) *Aug. 2025*
[Link] (in Chinese)

Conference talks

Measuring urban physical environments using image deep features *Nanchang, China*
The 26th Inter-University Symposium on Asian Megacities *Sep. 2025*

Comprehensive urban space representation with varying numbers of street-Level images *Honolulu, US*
American Association of Geographers 2024 Annual Meeting *Apr. 2024*

Quantifying the bias in place emotion extracted from photos on social networking sites: A case study on a university campus *Beijing, China (online)*
2021 International Graduate Workshop on GeoInformatics *Dec. 2021*

Teaching

Conceptual Modelling and Programming *University of Vienna*
Lecture *Spring 2026*

Spatial Analysis and GIS Applications *Peking University*
Teaching assistant *Spring 2025*

Community Services

Journal Reviewer

- ISPRS Journal of Photogrammetry and Remote Sensing
- Landscape and Urban Planning
- Cities
- Computers, Environment and Urban Systems
- Building and Environment
- Geography and Sustainability
- Urban Foresty & Urban Greenning
- International Journal of Geographical Information Science
- Travel Behaviour and Society
- Applied Geography
- Scientific Reports
- IEEE Transactions on Human-Machine Systems
- Journal of Urban Technology
- Computational Urban Science
- Journal of Transportation Engineering and Information (in Chinese)

Grants

2025-2029	**Key Program of the National Natural Science Foundation of China (No. 42430106)** Spatio-temporal optimization with geographical artificial intelligence	*Contributor*
2024-2026	**National Natural Science Foundation of China (No. 42371468)** Representing Urban Physical Environment using Street View Imagery	*Contributor*
2022-2025	**National Natural Science Foundation of China-Excellent Young Scientists Fund Program (Overseas) (No. 41901321)** Geospatial Data Mining & Visual Intelligence	*Contributor*
2022-2023	**International Research Center of Big Data for Sustainable Development Goals (CBAS)** Key Technologies and Applications of Big Earth Data for Sustainable Development Goals	*Contributor*
2021-2022	**Tencent Academic Research Fund (No. RAGR20210101)** Identifying urban informal settlement using multi-source big geo-data	*Key Contributor*
2020-2022	**National Natural Science Foundation of China (No. 41901321)** Understanding the relationship between urban physical environment and urban mobility patterns quantitatively using street view imagery	*Contributor*
2017-2021	**National Basic Research Program of China (No. 2017YFB0503602)** Geospatial Big Data Mining and Spatiotemporal Pattern Discovery	*Contributor*

Skills

Programming	Python, Matlab, R, Javascript, LaTex
Packages	Sklearn, Pytorch
Databases	MySQL, SQLite, MongoDB
GIS skills	ArcGIS, QGIS, Geopandas, Arcpy
OS	Windows, Linux, Mac OS
Languages	Mandarin Chinese *(Native)*, English *(Advanced)*